

Week 05: Type classes and instances

Functional Programming in Lean

Learning goals

- ▶ Read a polymorphic class constraint
- ▶ Define one class and two instances
- ▶ Trace instance selection at a call site

Central idea

A class describes an operation available for several types.

Lean example

```
class Label ( $\alpha$  : Type) where  
  label :  $\alpha \rightarrow \text{String}$ 
```

```
def describe [Label  $\alpha$ ] (x :  $\alpha$ ) : String :=  
  Label.label x
```

What to notice

- ▶ A class is an interface of operations
- ▶ An instance supplies those operations for a type
- ▶ The argument in square brackets is resolved by instance search

Live coding

1. Write the `TrafficLight` instance
2. Call `describe` on a light and a natural number
3. Inspect the inferred type with `#check`

Pause and predict

Where does Lean obtain `Label TrafficLight`?

Exercise connection

Open `exercises/Week05.md` after the lecture.

The complete example is `Course/Week05.lean`; the fill-in version is `Templates/Week05.lean`.